

Overview

We present the decision-making module of a low-cost social robot [1–3] for child Autism Spectrum Disorder (ASD) therapy.

- Since ASD is associated with difficulties in social communication, social robots have emerged as promising tools to support therapeutic interventions with children [4].
- The system follows a closed loop: **sensing** the child, **deciding** the robot response, and **acting** through an engaging behavior.
- The interaction is **multimodal**, integrating vision, audio, touch, and inertial sensing with speech, facial expressions, movement, and LEDs.
- The module uses an **expected-utility framework** to balance engagement, communication support, and therapeutic progress while accounting for online learning.



Open Conversation Framework

The module is deployed in a free-form dialogue activity structured around therapeutic conversational topics built from expert knowledge.

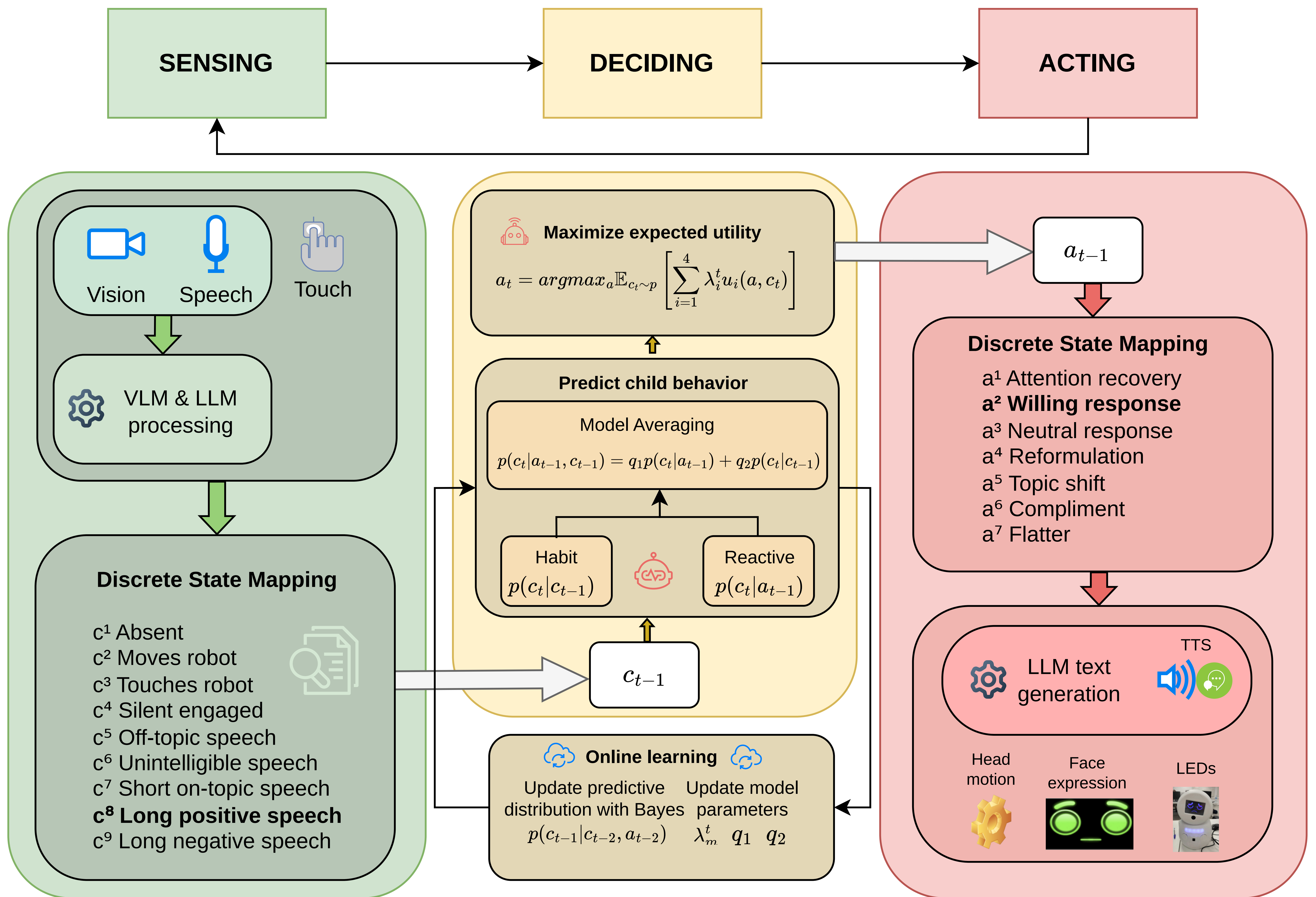
The interaction follows a finite-state loop: **SENSING** → **DECIDING** → **ACTING**.

- **SENSING**: the child is observed through speech, vision and physical interaction. Sensor data are mapped into a discrete child action $c_t \in \mathcal{C}$.
- **DECIDING**: the robot selects a response using a probabilistic decision-making module conditioned on the inferred child action.
- **ACTING**: the selected action is executed as a multimodal behaviour combining LLM-based speech, facial expression, head motion, and LED feedback.

Probabilistic Decision Making

- **Child-state prediction**: the robot combines a **habit** model of the child's subsequent behavior with a **reactive** model of the child's response to robot actions.
- **Expected utility maximization**: the robot chooses the action with highest expected utility.
- **Multiple objectives**: the utility, $U_t(a_t, c_t) = \sum_{m=1}^M \lambda_m^t u_m(a_t, c_t)$, combines therapeutic objectives such as **engagement**, **communication willingness**, and **interaction progress**.
- **Online adaptation**: the model is updated after each interaction.

Overview of the probabilistic decision-making loop

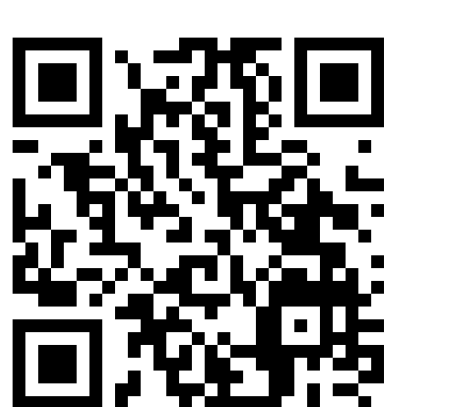


Conclusions

- Presented EMY's decision-making module for child ASD therapy, combining multimodal sensing and action with Dirichlet-based probabilistic modelling.
- The system adapts the robot's behaviour online based on the child's interaction.
- Future work will evaluate the framework in clinical settings.

References & Demo

- [1] S. Cooper *et al.*, "EMOROB CARE: A Low-Cost Social Robot for Supporting Children with Autism in Therapeutic Settings," *ICSR*, 2025.
- [2] S. Cooper *et al.*, "EMY: Supporting Autism Therapy with a Socially Assistive Robot," *ACM/IEEE HRI Companion*, 2026.
- [3] B. Pou and R. Ros, "Real-Time Vision for Socially Aware Robots: Gesture, Pointing, and Visual Engagement Estimation," *ACM/IEEE HRI Companion*, 2026.
- [4] P. Pennisi *et al.*, "Autism and social robotics: A systematic review," *Autism Research*, 2016.



Video demo